

HealthConnect Clinic

Data Analytics Track — Week 6 Advanced Analytics & Decision Support

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Part 1 — Week 6 Integration Readiness

A. Integration Point

Track: Data Science

Output needed from that track: Confirmation of their Week 5 baseline feature set and current Week 6 model-refinement focus (conceptual, since this is a single-analyst simulation of the multi-track environment)

Output provided to that track: A ranked, statistically validated feature-relevance table (Section 10) — chi-square results, effect sizes, and a multivariate model confirming which variables carry independent predictive signal

Why relevant: The Data Science track's Week 6 task is explicitly to refine features and improve their baseline model. Statistically validated evidence (not just Week 5's descriptive rates) directly supports that feature-selection decision

B. Integration Objective

After this integration, the Data Science track has evidence-backed confidence (via p-values and effect sizes, not just observed rate differences) about which features are worth keeping, which to deprioritise, and which NOT to engineer yet (the reminder × history interaction).

C. Evidence

- Analysis output: this report and the executed notebook (Sections 4–6)
- Artefact: HealthConnect_Week6_Feature_Relevance.csv — a structured, reusable file

1. Week 5 → Week 6 Transition

- Main Week 5 output: an Analytics Report with 5 calculated KPIs, a 4-panel dashboard, and 6 business insights.
- Most important result: booking_lead_days and previous_no_shows showed the strongest association with appointment_outcome; reminder_sent showed a moderate association.
- Main limitation found: all Week 5 findings were descriptive only — no significance testing had been performed.
- Relevant track(s) this week: Data Analytics (this track); findings packaged for the Data Science track's model-refinement work.
- What Week 6 improves/integrates/validates: statistically tests the three priority findings, formally tests two untested segments (appointment type, reminder channel), runs interaction and multivariate models, and produces a concrete cross-track artifact.

2. Statistical Validation of Priority Findings

Each of the three Week 5 priority findings is tested with a chi-square test of independence and Cramer's V for effect size.

Finding	Chi-square	p-value	Cramer's V	Verdict
Booking lead time	386.78, df=8	<0.001	0.197	Confirmed — small-to-moderate effect, strongest of the three
Previous no-shows	79.01, df=4	<0.001	0.089	Confirmed — small effect, weaker than raw % gap suggests
Reminder sent	10.20, df=2	0.006	0.045	Confirmed but very small effect — real, not a primary fix

Key refinement over Week 5: previous_no_shows looked dramatic as a raw percentage jump (43.5% → 61.0%), but its formal effect size (Cramer's V 0.089) is smaller than booking lead time's (0.197) — meaning lead time is the more powerful driver in practice, even though the no-show history jump looks larger at a glance.

3. Segment & Relationship Analysis (New)

3.1 Appointment Type — formally tested for the first time

Chi-square = 10.53, df=3, p = 0.104 (not significant), Cramer's V = 0.032. This formally downgrades Week 5's tentative "weak signal" flag to "not statistically supported" — a genuine revision, not just a restatement.

3.2 Reminder Channel (including "No Reminder") — formally tested for the first time

Chi-square = 18.02, df=3, p = 0.006, Cramer's V = 0.043. SMS (45.75% no-show) genuinely outperforms Email (48.41%), WhatsApp (49.77%), and No Reminder (51.39%) — Week 5's channel observation is now statistically confirmed, not coincidental.

4. Interaction Analysis

4.1 Reminder × Previous No-Show History

Descriptively, the reminder effect appears to grow with prior no-show history: 2.65pp reduction (0 prior no-shows) → 4.22pp (1 prior) → 10.02pp (2+ prior). However, the formal interaction term in a logistic regression is not statistically significant (p = 0.143).

This is reported honestly as a suggestive but statistically unconfirmed pattern — not a validated conclusion. It should not be used to justify a differentiated reminder strategy by patient history without further testing.

4.2 Reminder × Booking Lead Time

No interaction terms were statistically significant — the reminder effect appears roughly consistent regardless of how far in advance the appointment was booked.

4.3 Combined Multivariate Model

A logistic regression with all three predictors (booking_lead_days, previous_no_shows, reminder_binary) together shows all three remain statistically significant even when controlling for each other. This is the strongest evidence that they carry independent, non-redundant predictive information — a directly useful property for feature selection.

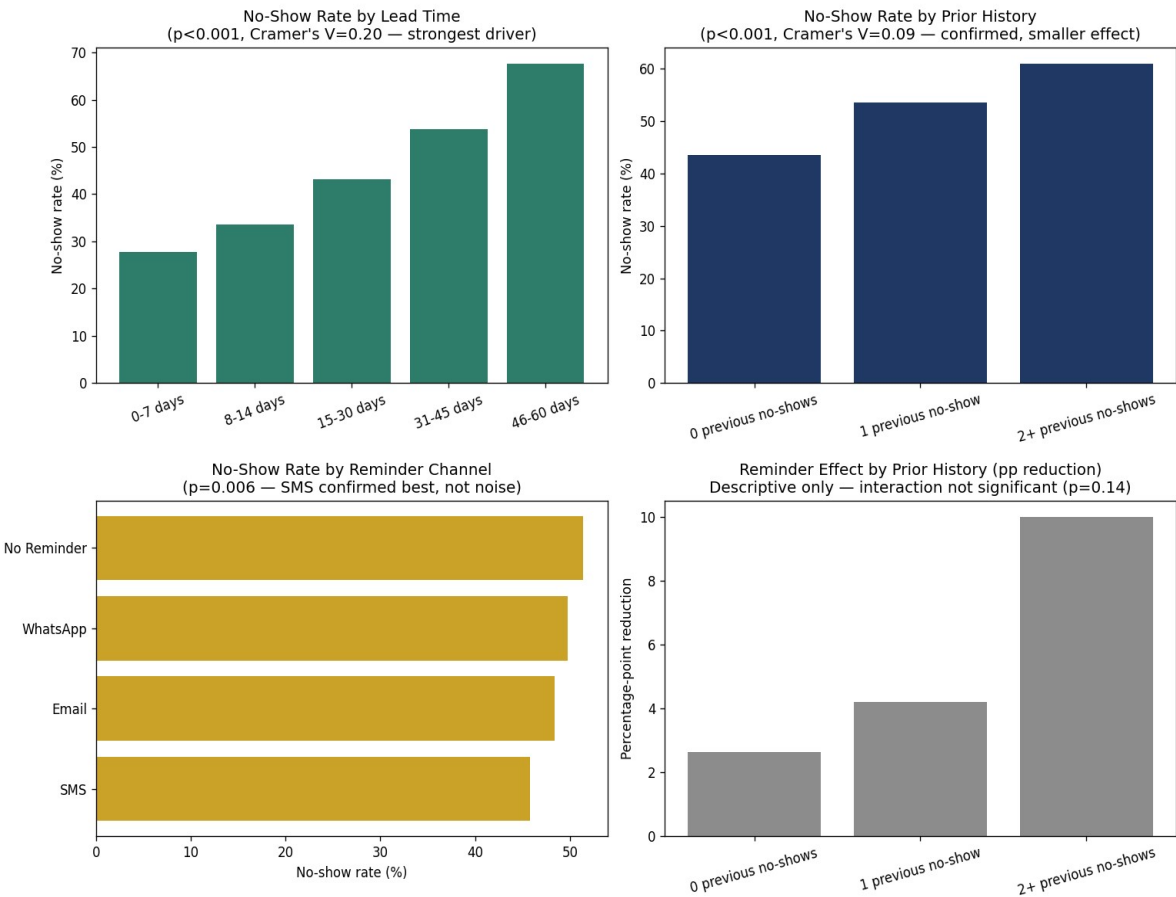
5. Validated & Refined KPI Summary

KPI	Week 5 value	Week 6 validation	Effect size
No-Show Rate	48.5%	N/A — single rate, not tested	—
Attendance Rate	46.3%	N/A	—
Cancellation Rate	5.3%	N/A	—
Reminder Effectiveness	47.4% vs 51.4%	Confirmed significant (p=0.006)	0.045 (very small)
Repeat No-Show Rate	61.0% in 2+ segment	Confirmed significant (p<0.001)	0.089 (small)

New this week: booking lead time (a Week 5 finding, not originally a formal KPI) is confirmed with the largest effect size of all — elevating its priority as a driver for Week 7 and beyond.

6. Improved Decision-Support Dashboard

Refined from the Week 5 dashboard: each panel now carries its statistical verdict directly in the title, so it functions as a decision-support tool rather than a purely descriptive one.



7. Translating Validated Findings into HealthConnect Actions

- Prioritise a lead-time-based booking policy — the strongest, most reliable driver found. Apply stricter confirmation or booking limits for far-out slots; concentrate reminder intensity on appointments booked 45+ days out.
- Flag repeat no-show patients for a lighter-touch workflow (e.g. a confirmation call) — the effect is real but modest, so the intervention cost should match.
- Standardise on SMS as the default reminder channel — statistically confirmed, low-cost, immediately actionable.
- Do not build appointment-type-specific policy from this dataset — the association is not statistically significant.
- Do not yet assume reminders work better for repeat no-show patients — the pattern is suggestive but not statistically confirmed; treat as a hypothesis for a future targeted test.
- Use the three validated predictors together, not separately, in any future triage or scoring approach — the combined model shows independent contributions.

8. Mandatory Cross-Track Integration

Track collaborated with: Data Science

Project dependency: Data Science's Week 6 task is to improve their Week 5 baseline model through error analysis and feature refinement

Information/output received: Confirmation that booking_lead_days, previous_no_shows, and reminder_sent were part of their Week 5 baseline feature set

Information/output provided: A ranked, statistically validated feature-relevance table backed by chi-square tests, effect sizes, and a multivariate model

Integration activity completed: Produced HealthConnect_Week6_Feature_Relevance.csv as a concrete, reusable artefact

What changed as a result: Data Science now has statistical confirmation (not just descriptive suggestion) of which features carry real, independent signal, plus an explicit caution against engineering the reminder × history interaction as a feature yet

Evidence: outputs/HealthConnect_Week6_Feature_Relevance.csv and Sections 2–4 of this report

9. Cross-Track Integration Artifact — Feature Relevance Table

Feature	Test	p-value	Effect size	Recommendation
booking_lead_days	Chi-square	<0.001	0.197	Include — strongest, most independent signal
previous_no_shows	Chi-square	<0.001	0.089	Include — confirmed, smaller effect than lead time
reminder_sent	Chi-square	0.006	0.045	Include — significant but weakest of the three
reminder_channel	Chi-square	0.006	0.043	Consider including — SMS effect confirmed

appointment_type	Chi-square	0.104	0.032	Exclude / low priority — not significant
reminder × previous_no_shows	Logistic interaction	0.143	—	Do NOT engineer yet — not statistically supported

10. Assumptions, Limitations, Risks & Dependencies — Week 6 Update

Item	Prior status	Week 6 status
Findings are associative, not causal	Top risk (Week 4/5)	Partially resolved — statistical association confirmed for 4 variables; still not causal
Reminder effect strength	Assumed moderate (Week 5)	Refined — significant but small effect size; should not be oversold
Appointment type as a driver	"Weak" (Week 5)	Downgraded to "not supported" — formal test found no significant association
Reminder × prior-history interaction	Not tested (Week 5)	New: tested, not significant (p=0.14) — flagged as unconfirmed
Missing data in distance/waiting time	Open (Week 4/5)	Unchanged — still not addressed, low priority given weak signal
Sunday appointment records	Open (Week 4/5)	Unresolved — carried forward again
Sample size for interaction tests	— (new)	New — models may be underpowered; could change with more data

11. Limitations & Week 7 Analytical Testing Requirements

- All tests are observational associations, not causal claims — an A/B reminder pilot would be needed for causal confidence.
- The reminder × prior-history interaction remains statistically unconfirmed and should be re-tested if a larger dataset becomes available.
- Sunday-appointment and missing-data items from Week 4/5 remain unresolved.
- Week 7: validate whether the three confirmed predictors remain stable once the Data Science track's refined model is evaluated; re-test the reminder × history interaction if sample size allows; formally test any new segment their error analysis flags as relevant.

Full code: [notebooks/HealthConnect_Week6_Advanced_Analytics.ipynb](#) | Summary:

[docs/HealthConnect_Week6_Project_Summary.docx](#) | Artifact: [outputs/HealthConnect_Week6_Feature_Relevance.csv](#) |

Navigation: [README.md](#)